



Local NMBU storage



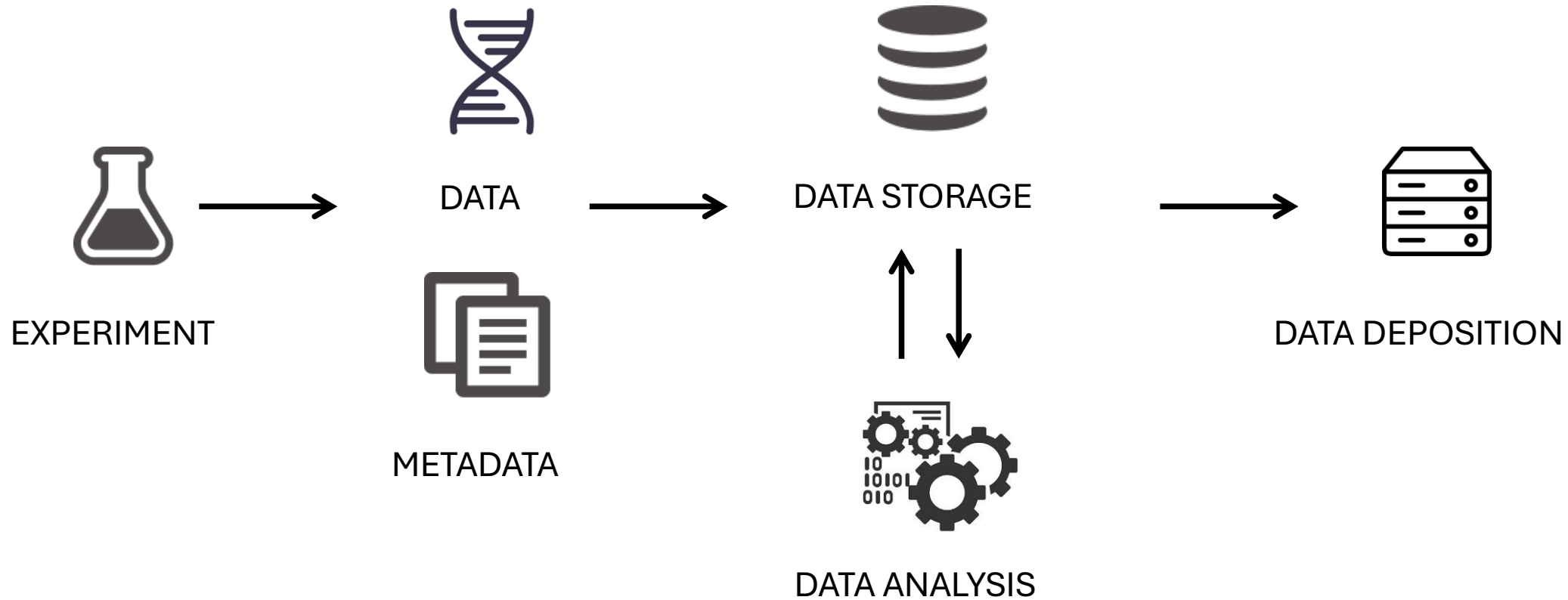
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ELIXIR Norway



Why Data Management Plan?

- Requirements from the funding agencies (NFR, EU)
- From July 2020 - all new PhD candidates at NMBU should have a Data Management Plan (DMP)
- NMBU Guidelines for Research Ethics

Data storage – from the researcher perspective



Storage of Research Data without personal information - tools

- **Research Data** generated in ongoing projects should normally be stored at the LargeFile server (W: or P:).
- **External cooperation (should not be used as the primary place for storage):** Sharing data using Office365 and OneDrive – NMBU services

LargeFile
server W:



Where to storage all these data?

NMBU has a set of file directory / areas for different proposes:

Home directory: OneDrive for Business - Norwegian University of Life Sciences (~ 1Tb)



Common directory (Y:) It serves to share data with intra faculty members

LargeFile area (W:)

Area for storing large amounts of data (> 1Tb)

➤  Research Home (J:)

➤  Research Project (P:)

➤  Research Home (W:)

~~➤  Public (Y:)~~

NMBU Full-managed client.

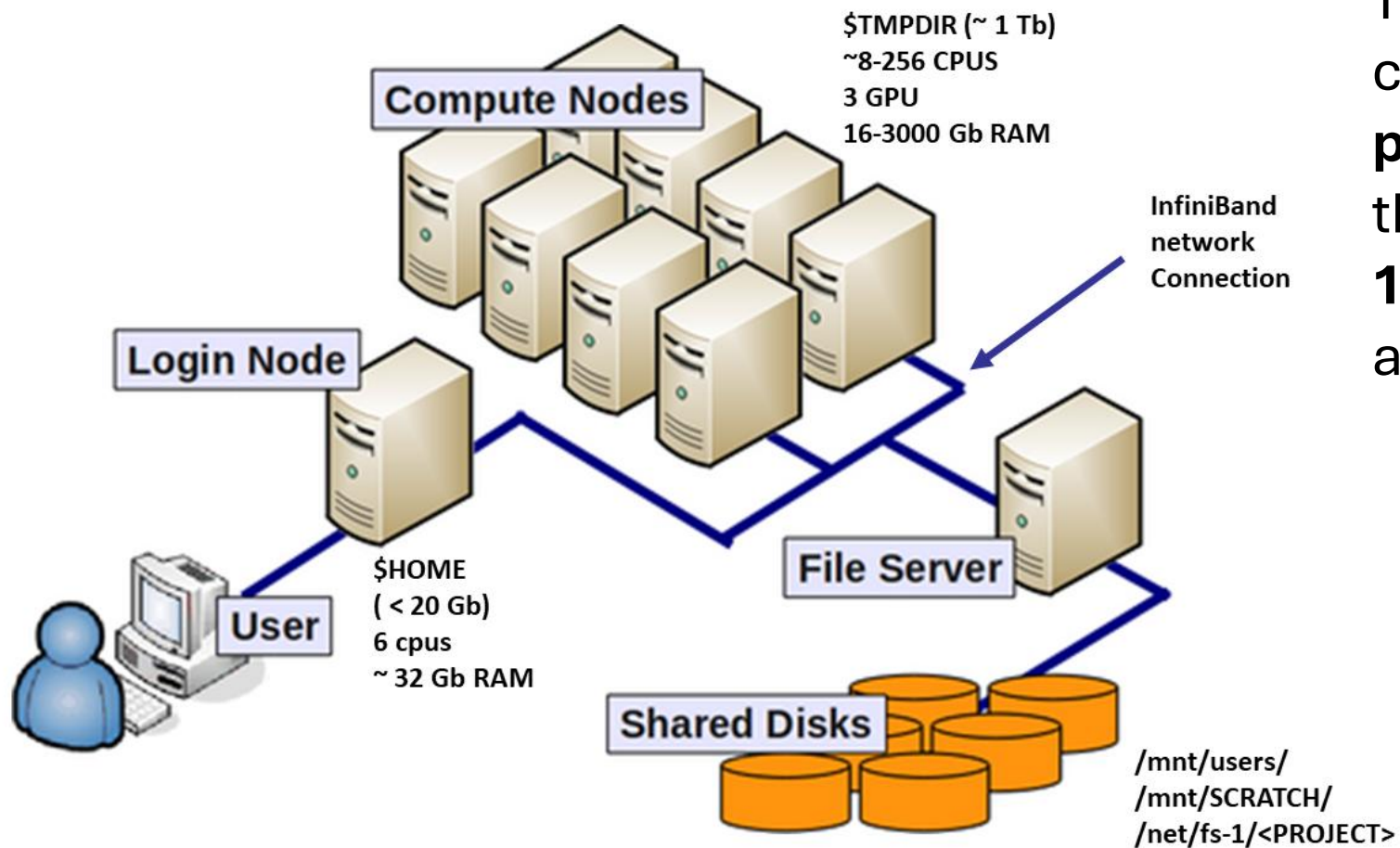
LabFile area (P:)

Area for storing data from Lab Machines.

<https://nmbuhjelp.nmbu.no/>

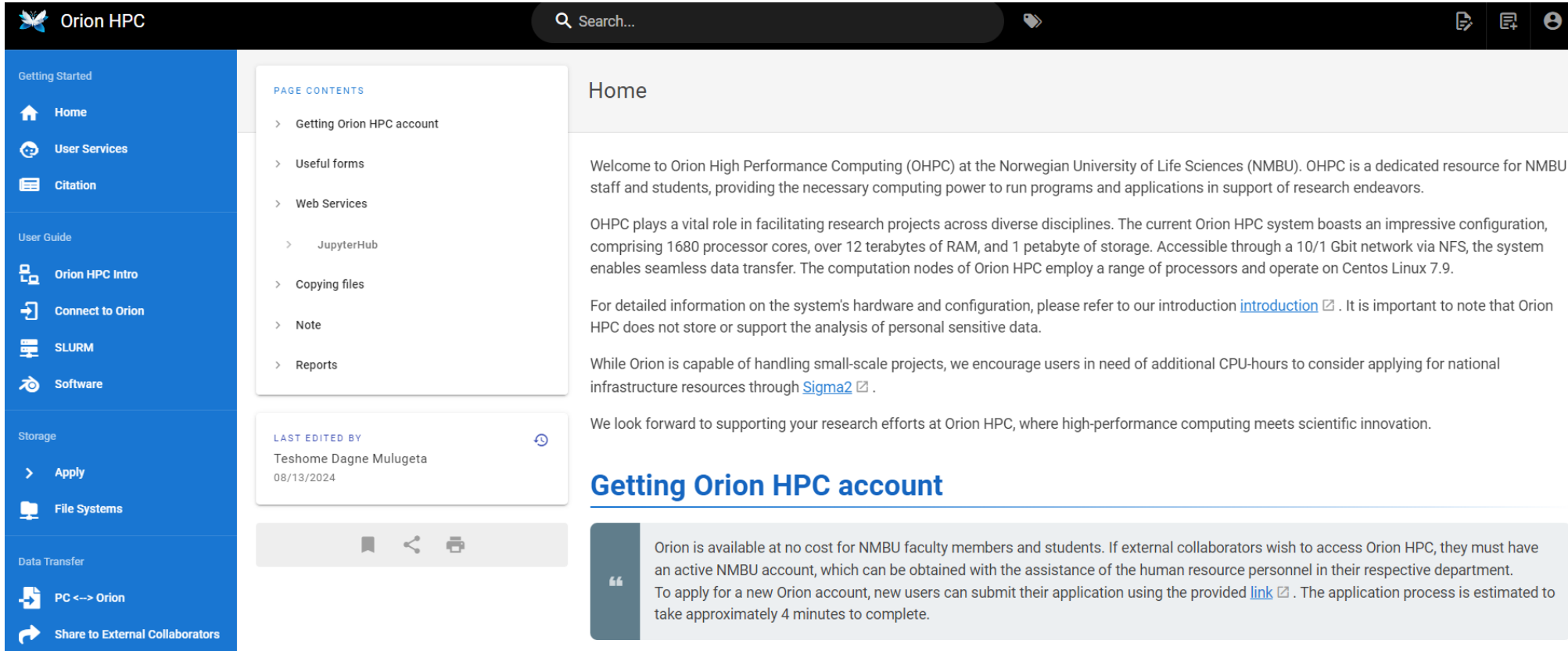
Datamigrering til organisatoriske team

Access to the NMBU High-Performance Computing (HPC) grid Orion



The Orion HPC system currently consists of **1680 processor cores** with more than **12 terabytes of RAM** and **1 Pb of storage** accessible on a 10/1 Gbit network via NFS.

How can I get access to Orion?



The screenshot shows the Orion HPC website. The top navigation bar includes the Orion HPC logo, a search bar, and icons for document, list, and user. The left sidebar is divided into sections: 'Getting Started' (Home, User Services, Citation), 'User Guide' (Orion HPC Intro, Connect to Orion, SLURM, Software), 'Storage' (Apply, File Systems), and 'Data Transfer' (PC <--> Orion, Share to External Collaborators). The main content area is titled 'Home' and contains a 'PAGE CONTENTS' sidebar with links to 'Getting Orion HPC account', 'Useful forms', 'Web Services', 'JupyterHub', 'Copying files', 'Note', and 'Reports'. The main text area welcomes users to Orion High Performance Computing (OHPC) at NMBU, describes the system's capabilities (1680 processor cores, 12 terabytes of RAM, 1 petabyte of storage), and provides information on how to get an account. A 'Getting Orion HPC account' section highlights that access is free for NMBU faculty and students, and provides a link for external collaborators. A QR code is located in the bottom right corner of the page.

Orion HPC

Search...

Getting Started

- Home
- User Services
- Citation

User Guide

- Orion HPC Intro
- Connect to Orion
- SLURM
- Software

Storage

- Apply
- File Systems

Data Transfer

- PC <--> Orion
- Share to External Collaborators

PAGE CONTENTS

- Getting Orion HPC account
- Useful forms
- Web Services
- JupyterHub
- Copying files
- Note
- Reports

Home

Welcome to Orion High Performance Computing (OHPC) at the Norwegian University of Life Sciences (NMBU). OHPC is a dedicated resource for NMBU staff and students, providing the necessary computing power to run programs and applications in support of research endeavors.

OHPC plays a vital role in facilitating research projects across diverse disciplines. The current Orion HPC system boasts an impressive configuration, comprising 1680 processor cores, over 12 terabytes of RAM, and 1 petabyte of storage. Accessible through a 10/1 Gbit network via NFS, the system enables seamless data transfer. The computation nodes of Orion HPC employ a range of processors and operate on Centos Linux 7.9.

For detailed information on the system's hardware and configuration, please refer to our introduction [introduction](#). It is important to note that Orion HPC does not store or support the analysis of personal sensitive data.

While Orion is capable of handling small-scale projects, we encourage users in need of additional CPU-hours to consider applying for national infrastructure resources through [Sigma2](#).

We look forward to supporting your research efforts at Orion HPC, where high-performance computing meets scientific innovation.

Getting Orion HPC account

Orion is available at no cost for NMBU faculty members and students. If external collaborators wish to access Orion HPC, they must have an active NMBU account, which can be obtained with the assistance of the human resource personnel in their respective department. To apply for a new Orion account, new users can submit their application using the provided [link](#). The application process is estimated to take approximately 4 minutes to complete.

<https://orion.nmbu.no/>

Free for all NMBU community 😊



Virtual GUI services Orion offers

Orion OnDemand

Files

Jobs

Clusters

Interactive Apps

Tools

My Interactive Sessions

Clusters / Access

Home Directory

Browse and manage files in your home directory.

Open Home Directory

Orion Shell Access

Open a shell on login nodes.

Open Shell

Interactive Apps

RStudio Server

Launch an RStudio Server session.

Launch RStudio

Jupyter Server

Start a Jupyter notebook server.

Launch Jupyter

Home / My Interactive Sessions

Interactive Apps

Desktops

Orion Desktop

Servers

Orion Jupyter

Orion RStudio

Orion RStudio (1029565)

1 node | 2 cores | Running

Host: [Dm-cs-36](#)

Created at: 2026-05-22 12:03:58 CEST

Time Remaining: 4 hours and 56 minutes

Session ID: 156b8618-b9ec-4e37-a23b-f769b61a0f3

Module issues in terminal

There's a known issue loading modules in RStudio's environment after changing versions or clusters.

If you have issues using modules in the RConsole - try these remedies

- restarting the terminal
- restarting the RConsole
- logging out of the RStudio session and logging back in.
- remove your ~/.local/share/rstudio

Contact Orion Support if you continue to have issues.

How to import modules in R

Connect to RStudio Server

Orion RStudio (1029564)

1 node | 2 cores | Running

Host: [Dm-cs-36](#)

Created at: 2026-05-22 11:48:38 CEST

Time Remaining: 4 hours and 41 minutes

Session ID: b1b0e692-c46e-423c-9f0d-eb2e9145ccac

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How to import modules in R

Connect to RStudio Server

Home / My Interactive Sessions / Orion RStudio

Interactive Apps

Desktops

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Orion RStudio

Orion RStudio

This app will launch RStudio Server an IDE for R on the Orion clusters.

RStudio / R version

R 4.4.2 (RStudio in Apptainer)

R 4.5.3 (RStudio in Apptainer)

R 4.4.2 (RStudio in Apptainer)

Number of hours

5

Number of cores

2

Minimums and Maximums will change depending on your choice of Node type and Cluster.

Memory (GB)

20

Total RAM reserved for the RStudio job.

☐ I would like to receive an email when the session starts

Launch

* The Orion RStudio session data for this session can be accessed under the `data root` directory.



Figure: <https://www.swisslife.com/en/home/blog/fuck-you-money.html>

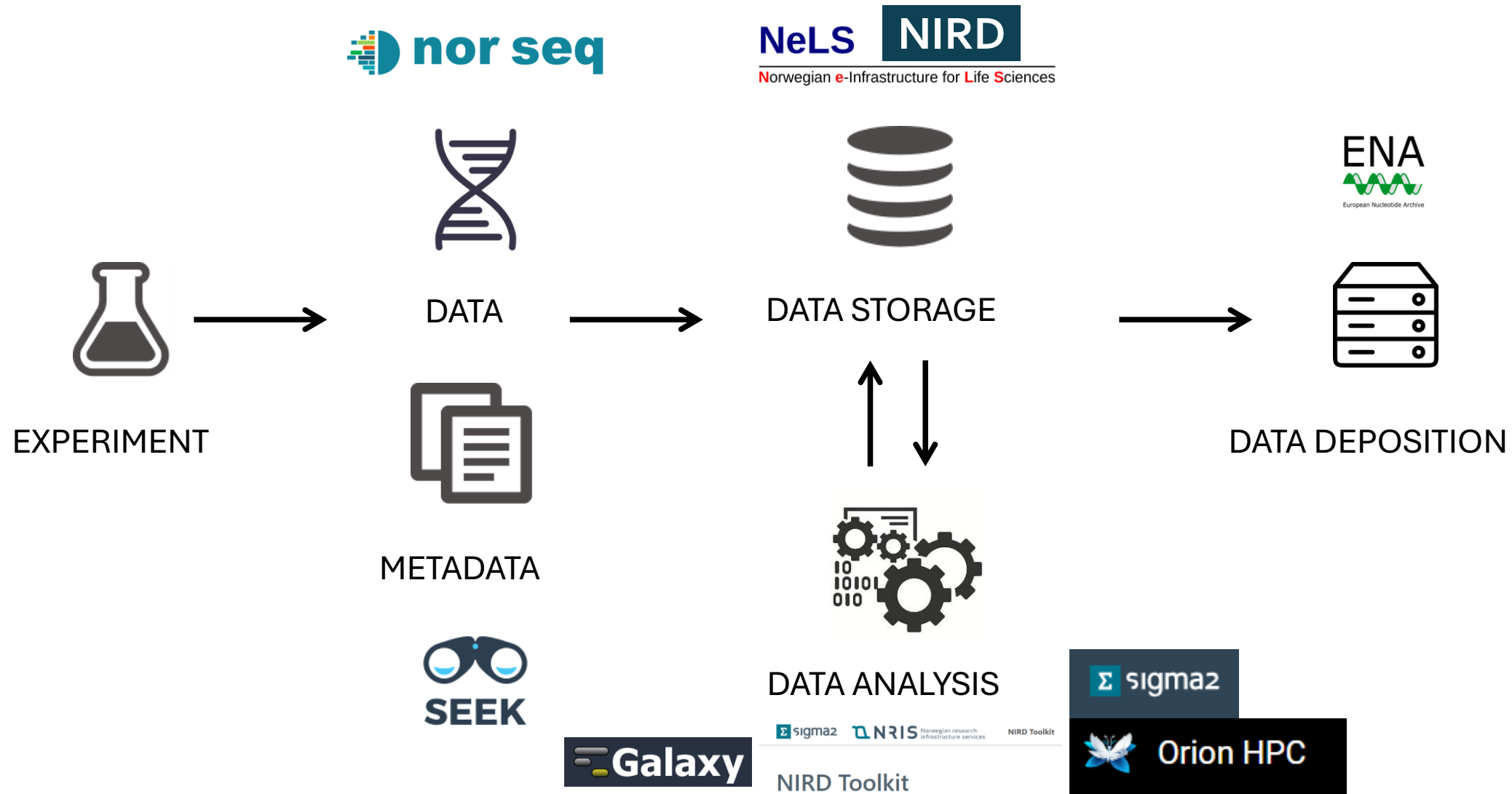
Usage price

- Leiested model
 - 0.072 per core hour
 - 1,383 per TB/year

Nothing in life
is free... Costs:

- Applies from July/August 2026 for new projects
- January 2027 for everyone using Orion

Example - Data flow/handle using ELIXIR Norway



Summary NMBU guidelines



- The project leader have filled in a Data Management Plan (DMP) at the start of a project (continuous process)
- Data is stored at NMBU server or at national/ international platforms in ongoing projects.
- Data is tagged with metadata.
- Data is deposited in archives at the end of a project/ publication.
- Data is made open and available when/ if possible.



- Data is only stored at the local hard drive (C: disk).
- Data is only stored at Office365
- Data is only stored at an external hard drive.
- Data is stored in Dropbox.